

EE 474 Introduction to Control Systems**Homework 5**

Due Wednesday, October 14, 2026

Announcements :

- **The midterm will be Monday, October 19.**
 - The exam is closed-book, closed-notes, and most electronic devices are prohibited.
 - These items are allowed:
 - * **Calculator:** A non-programmable scientific calculator is allowed, such as a TI-30Xa, available at Walmart (\$8.94).
 - No other electronic devices are allowed, specifically, no cell phone, smart watch, smart glasses, communicating device or computer may be out during the exam.
 - You will not need a calculator, but some people prefer to have one. A non-programmable calculator is allowed (such as a TI-30Xa, \$11.18 at Walmart).
 - NO PROGRAMMABLE CALCULATOR (can give an unfair advantage)
 - * **Notes:** You are invited to review material covered in the course, including readings, lecture and homework assignments, and prepare one 8.5" by 11" sheet of notes (2-sided). You may use your sheet of notes during the exam. (Two pages of notes will be allowed on the final).
 - I'll post a past exam, to give an idea of what the exam will be like.

Reading:

- Franklin *et al.*: Remainder of chapter 3, except for sections 3.6.3, 3.7 and 3.8 (7th or 8th ed.)

Homework :**1 Transfer Functions :**

A) For a system described by the block diagram of figure 1

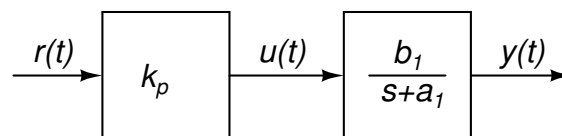


Figure 1: Block diagram.

Find the transfer function

$$G_{ry1}(s) = \frac{Y(s)}{R(s)} \quad (1)$$

B) For a system described by the block diagram of figure 2 :

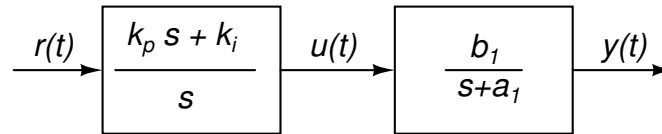


Figure 2: Block diagram.

Find the Transfer function

$$G_{ry2}(s) = \frac{Y(s)}{R(s)} \quad (2)$$

C) For a system described by the block diagram :

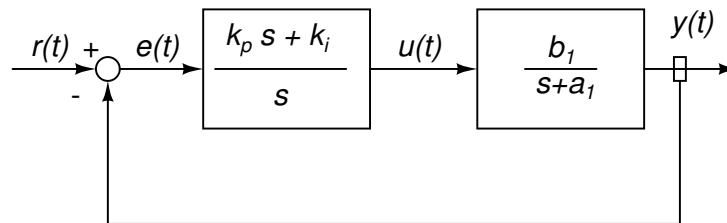


Figure 3: Block diagram.

C.1) Find the Transfer function

$$T_{ry3}(s) = \frac{Y(s)}{R(s)} \quad (3)$$

C.2) Determine $T_{ry3}(s)$ for $b_1 = 3$, $a_1 = 2$, $k_p = 1$, $k_i = 5$.

C.3) What are the pole and zero locations for these parameters

C.4) Plot the step response of the system of figure 3 for the parameters given.

C.5) From the plot, how great is the overshoot ?

(Overshoot is usually expressed as a percentage of the steady state value).

C.6) The controls engineer needs to limit the P.O. to 5%, and changes k_p (a controller parameter) to $k_p = 1.6$. Is the overshoot less than 5% ?

((There are a variety of ways to explore matlab plots, including finding the magnitude of the peak. One way is to hover with the cursor over a point on a curve. Another is :

```
[yy, tt] = step( Try3 ); max(yy)
```

The first command returns the output and time values, the second finds the peak.))

2 For transfer functions $G_1(s)$ to $G_5(s)$, do or determine :

- A) If the system is stable
- B) Hand sketch a plot of the pole constellation (suggestion: use Matlab to determine where the pole and zero values).
- C) If the system is stable and has a complex pole pair, determine the damping factor.

Note: if pole location $p_1 = -\alpha + j\omega$, then the damping factor is given as

$$\rho = \alpha/\omega_n \quad \text{with} \quad \omega_n^2 = \alpha^2 + \omega^2,$$

$$G_1(s) = \frac{s+2}{s^2+10s+8}$$

$$G_2(s) = \frac{5s+15}{s^2+2s+10}$$

$$G_3(s) = \frac{s+3}{s^2-4s+8}$$

$$G_4(s) = \frac{5s+15}{s^3+2s^2+6s+10}$$

$$G_5(s) = \frac{2s+9}{s^3+2s^2+6s+30}$$

3 Partial Fraction Expansion

Text Problems 3.7(a) and 3.7(c), by any 2 of these methods :

- A) Plug in and solve
- B) Cover-up
- C) Graphical

Text problem 3.7 : Find the time function corresponding to each of the following Laplace transforms using partial-fraction expansions:

(a)

$$F(s) = \frac{1}{s(s+1)}$$

(c)

$$F(s) = \frac{3s+2}{s^2+2s+10}$$